RIS / RECONFIGURABLE METASURFACE STUDY PLAN

For: 3rd-year ELEL student, CityU

Date prepared: 2026-06-21

Goal: Build enough background to understand the provided references on mmWave, RIS, reflectarrays, tunable metasurfaces, IR-controlled programmable surfaces, and RIS optimization.

A layered reading plan for building the background needed to understand mmWave, RIS, reflectarrays, tunable metasurfaces, IR-controlled programmable surfaces, and RIS optimization.

0. How To Use This Plan

Do not read the references in numerical order. Your references combine several different layers:

1) Electromagnetic waves
2) Microwave / RF circuits
3) Antennas and arrays
4) Microstrip antennas and reflectarrays
5) Metamaterials and metasurfaces
6) Wireless communication theory
7) RIS communication models and optimization
8) Tunable hardware: PIN diodes, varactors, optical/IR control
9) Machine learning / optimization for programmable metasurfaces

The recommended order is:

EM waves -> microwave engineering -> antennas/arrays -> microstrip/reflectarrays -> metasurfaces -> wireless/RIS communication -> hardware tuning/control -> optimization/ML.

You do not need to master every chapter before reading papers. The aim is to understand enough vocabulary, equations, and physical intuition so that the papers become readable.

1. 12-Week Reading Plan

WEEK 1: Electromagnetic waves and reflection basics

Main book:

- Sophocles J. Orfanidis, Electromagnetic Waves and Antennas

Read:

- Chapter 1: Maxwell's Equations
- Chapter 2: Uniform Plane Waves
- Chapter 5: Reflection and Transmission

Goal:
Understand how waves propagate, what electric/magnetic fields are doing, what reflection means, and why boundary conditions matter.

Key ideas:

- Maxwell equations in phasor form
- Plane waves
- Wave impedance
- Reflection coefficient
- Transmission coefficient
- Boundary conditions

Why this matters:
RIS and metasurfaces are about controlling how EM waves reflect, transmit, and scatter.

WEEK 2: Oblique incidence and transmission lines

Main books:

- Orfanidis, Electromagnetic Waves and Antennas
- David M. Pozar, Microwave Engineering

Read:

- Orfanidis Chapter 7: Oblique Incidence
- Orfanidis Chapter 11: Transmission Lines
- Pozar Chapter 2: Transmission Line Theory

Goal:

Understand why angle of incidence matters and why transmission-line thinking is everywhere in microwave/RF design.

Key ideas:

- Normal vs oblique incidence
- TE/TM polarization
- Standing waves
- Characteristic impedance
- Reflection coefficient Gamma
- VSWR
- Input impedance

Why this matters:

Metasurface unit cells are often modeled as impedance sheets or resonant transmission-line-like elements.

WEEK 3: Microwave network analysis and matching

Main books:

- Pozar, Microwave Engineering
- Orfanidis, Electromagnetic Waves and Antennas

Read:

- Pozar Chapter 3: Transmission Lines and Waveguides

Focus especially on microstrip lines if your edition has that section.

- Pozar Chapter 4: Microwave Network Analysis
- Pozar Chapter 5: Impedance Matching and Tuning
- Orfanidis Chapter 13: Impedance Matching
- Orfanidis Chapter 14: S-Parameters

Goal:

Understand how RF engineers describe circuits at high frequency.

Key ideas:

- S11, S21, S12, S22
- Scattering matrix
- Return loss
- Insertion loss

- Smith chart
- Matching networks
- Microstrip lines
- Waveguides

Why this matters:

RIS/metasurface papers often show reflection phase, reflection magnitude, S11, and phase response of a unit cell.

WEEK 4: Antenna basics and array factor

Main books:

- Constantine A. Balanis, Antenna Theory: Analysis and Design
- Orfanidis, Electromagnetic Waves and Antennas

Read:

- Balanis Chapter 2: Fundamental Parameters of Antennas
- Balanis Chapter 6: Arrays: Linear, Planar, and Circular
- Orfanidis Chapter 16: Transmitting and Receiving Antennas
- Orfanidis Chapter 22: Antenna Arrays

Goal:

Understand radiation patterns and beam steering.

Key ideas:

- Gain
- Directivity
- Radiation pattern
- Beamwidth
- Sidelobes
- Array factor
- Phase progression
- Beam steering
- Aperture size vs beamwidth

Why this matters:

RIS beam steering is basically phase control across a surface. Array factor is one of the easiest ways to understand it.

Mini task:

Use MATLAB or Python to plot the array factor of a linear array while changing the phase shift between elements.

WEEK 5: Microstrip antennas and reflectarray basics

Main books:

- Balanis, Antenna Theory: Analysis and Design
- Garg et al., Microstrip Antenna Design Handbook
- Nayeri, Yang, Elsherbeni, Reflectarray Antennas: Theory, Designs, and Applications

Read:

- Balanis Chapter 14: Microstrip Antennas

- Microstrip Antenna Design Handbook:
  - Chapter 1: overview, advantages, limitations, feeding, applications
  - Then read the chapters/sections on rectangular patches, feeding methods, matching, bandwidth, efficiency, and arrays.
  - Exact chapter numbering may vary by edition/printing, so follow these topic names if the chapter numbers differ.
- Reflectarray Antennas Chapter 1: Introduction
- Reflectarray Antennas Chapter 2: Reflectarray Element Design
- Reflectarray Antennas Chapter 3: Reflectarray System Analysis

Goal:
Understand how printed RF structures can radiate or reflect with controlled phase.

Key ideas:
- Patch resonance
- Substrate permittivity
- Effective dielectric constant
- Fringing fields
- Feed methods
- Reflection phase curve
- Unit cell
- Reflectarray aperture phase compensation

Why this matters:
Many RIS elements are close relatives of microstrip patches, reflectarray cells, or resonant printed scatterers.

WEEK 6: Reflectarray design, radiation patterns, bandwidth

Main books:
- Nayeri, Yang, Elsherbeni, Reflectarray Antennas
- Balanis, Antenna Theory: Analysis and Design

Read:
- Reflectarray Antennas Chapter 4: Reflectarray Radiation Pattern Calculation
- Reflectarray Antennas Chapter 5: Reflectarray Bandwidth
- Reflectarray Antennas Chapter 6: Reflectarray Design Examples
- Balanis Chapter 15: Reflector Antennas

Optional if your project involves beam scanning:
- Reflectarray Antennas Chapter 10: Beam Scanning Reflectarrays

Goal:
Understand classical reflectarrays before jumping into RIS.

Key ideas:
- Aperture phase distribution
- Path length compensation
- Phase quantization
- Reflectarray gain
- Aperture efficiency

- Bandwidth limitations
- Beam scanning

Why this matters:

RIS is often discussed in communications papers, but physically it is very close to a reconfigurable reflectarray/metasurface.

WEEK 7: Wireless communication basics for RIS

Main book:

- Andrea Goldsmith, Wireless Communications

Read:

- Goldsmith Chapter 2: Path Loss and Shadowing
- Goldsmith Chapter 3: Statistical Multipath Channel Models
- Goldsmith Chapter 10: Multiple Antennas and Space-Time Communications

Reference papers to read after this week:

- [25] ElMossallamy et al., Reconfigurable Intelligent Surfaces for Wireless Communications: Principles, Challenges, and Opportunities
- [4] Wu and Zhang, Towards Smart and Reconfigurable Environment
- [2] Basar et al., Wireless Communications Through Reconfigurable Intelligent Surfaces

Goal:

Understand what RIS means from a wireless-communication point of view.

Key ideas:

- Path loss
- Shadowing
- Multipath
- Fading
- Channel matrix
- MIMO
- Beamforming
- Passive beamforming
- Cascaded channel: transmitter -> RIS -> receiver

Why this matters:

Communication papers often write the RIS as a diagonal matrix of reflection coefficients. You need to know what that means physically and mathematically.

WEEK 8: RIS channel models, MIMO, OFDM, and cellular context

Main books:

- Goldsmith, Wireless Communications
- Tse and Viswanath, Fundamentals of Wireless Communication

Read from Goldsmith:

- Chapter 4: Capacity of Wireless Channels
- Chapter 7: Diversity

- Chapter 12: Multicarrier Modulation
- Chapter 15: Cellular Systems

Read from Tse and Viswanath as optional/advanced supplement:

- Chapter 2: The Wireless Channel
- Chapter 3: Point-to-Point Communication
- Chapter 5: Capacity of Wireless Channels
- Chapter 7: MIMO I: Spatial Multiplexing and Channel Modeling
- Chapter 8: MIMO II: Capacity and Multiplexing Architectures

Reference papers to read after this week:

- [3] Di Renzo et al., Smart Radio Environments Empowered by Reconfigurable Intelligent Surfaces
- [6] Tapio et al., Survey on Reconfigurable Intelligent Surfaces Below 10 GHz

Goal:

Understand the communication model deeply enough to follow RIS channel, beamforming, and performance-analysis papers.

Key ideas:

- Capacity
- Diversity
- OFDM
- MIMO channel modeling
- Channel state information, CSI
- Cellular propagation
- Near-field vs far-field when surface is large

Why this matters:

Many RIS papers are not only about hardware; they are about designing the environment as part of the wireless channel.

WEEK 9: Metamaterials and metasurfaces

Main sources:

- Ali, Mitra, Aissa, Metamaterials and Metasurfaces: A Review from the Perspectives of Materials, Mechanisms and Advanced Metadevices
- Glybovski et al., Metasurfaces: From Microwaves to Visible
- Tretyakov, Analytical Modeling in Applied Electromagnetics

Read:

- [16] Ali et al. first, as a broad review
- [11] Glybovski et al. second, as a deeper metasurface review
- Tretyakov:
  - Chapter 1: Introduction
  - Chapter 2: Thin Layers and Sheets
  - Then read the sections/chapters on periodic structures, artificial impedance surfaces, and metamaterials/metasurfaces if available in your edition.

Optional from Orfanidis:

- Chapter 6: Multiple Dielectric Slabs
- Chapter 10: Resonant Cavities

Goal:
Understand why a thin engineered surface can control phase, amplitude, polarization, absorption, and scattering.

Key ideas:
- Metamaterial
- Metasurface
- Surface impedance
- Sheet model
- Unit cell
- Periodic structure
- Subwavelength element
- Resonant scatterer
- Anomalous reflection/refraction

Why this matters:
RIS is the wireless-communications name for a broader physical idea: programmable control of wave propagation using engineered surfaces.

WEEK 10: Tunable and programmable RIS hardware

Main sources:
- Skyworks varactor diode application note
- Skyworks Design With PIN Diodes application note
- MACOM Design With PIN Diodes application note
- Pozar, Microwave Engineering

Read:
- Pozar Chapter 11: Active RF and Microwave Devices

Focus on diode behavior and RF device modeling.
- Application notes on:
  - Varactor diodes
  - PIN diodes
  - RF switches
  - Phase shifters

Reference papers to read after this week:
- [20] Hum et al., Modeling and Design of Electronically Tunable Reflectarrays
- [21] Sievenpiper et al., Two-Dimensional Beam Steering Using an Electrically Tunable Impedance Surface
- [22] Tian et al., 1-bit Digital Reconfigurable Reflective Metasurface
- [28] Kamoda et al., 60-GHz Electronically Reconfigurable Large Reflectarray
- [29] Yang et al., 1-bit Multipolarization Reflectarray Element
- [30] Sun et al., Infrared-Controlled Programmable Metasurface
- [10] Sayanskiy et al., 2D-Programmable and Scalable RIS Remotely Controlled via Digital Infrared Code

Goal:
Understand how a metasurface/RIS is physically reconfigured.

Key ideas:

- PIN diode as RF switch/current-controlled resistor
- Varactor diode as voltage-controlled capacitance
- Bias network
- RF choke
- DC blocking capacitor
- Quantized phase states: 1-bit, 2-bit, etc.
- Reflection phase tuning
- Control board / digital coding
- IR control of metasurface states

Why this matters:
A communications paper may write theta_n = exp(j phi_n), but real hardware must implement phi_n using physical tuning elements.

WEEK 11: Optimization and machine learning for RIS

Main book:
- Boyd and Vandenberghe, Convex Optimization

Read:
- Chapter 1: Introduction
- Chapter 2: Convex Sets, only the basics
- Chapter 3: Convex Functions, only the basics
- Chapter 4: Convex Optimization Problems
- Chapter 5: Duality, focus on Lagrangian and KKT ideas
- Chapter 6: Approximation and Fitting, focus on least squares
- Chapter 9: Unconstrained Minimization, focus on gradient/Newton/descent ideas

Optional:
- Skim Chapter 11: Interior-Point Methods only if a paper mentions semidefinite programming or barrier methods.

Reference paper to read after this week:
- [9] Zhang et al., Experiment-Based Deep Learning Approach for Power Allocation with a Programmable Metasurface

Goal:
Understand why RIS design becomes an optimization problem.

Key ideas:
- Objective function
- Constraints
- Convex vs nonconvex optimization
- Alternating optimization
- Gradient descent
- Least squares
- Lagrangian
- KKT conditions
- Semidefinite relaxation, SDR
- Discrete phase optimization
- Power allocation

Why this matters:

RIS optimization is hard because the surface is passive, phase-only or quantized, and coupled with the transmitter/receiver channel.

WEEK 12: mmWave, final integration, and paper reading

Main sources:

- Rappaport et al., Millimeter Wave Mobile Communications for 5G Cellular: It Will Work!
- Rappaport et al., Millimeter Wave Wireless Communications, if available
- Pozar Chapter 14: Introduction to Microwave Systems
- Goldsmith Chapter 15: Cellular Systems

Read:

- [1] Rappaport et al., Millimeter Wave Mobile Communications for 5G Cellular
- Revisit [3], [10], [11], [25]

Goal:

Connect mmWave propagation, antenna arrays, and RIS/metasurfaces.

Key ideas:

- mmWave path loss
- Blockage
- Beamforming
- Large arrays
- 28 GHz / 60 GHz systems
- High-gain directional links
- RIS as coverage/propagation control

Final task:

Write a 2-3 page summary answering:

1) What problem does RIS solve?
2) How does a unit cell change reflection phase?
3) How is RIS different from a normal antenna array?
4) What are the main hardware limitations?
5) What are the main communication-theory challenges?
6) Which of your references are about physics, which about hardware, and which about communication theory?

2. Book-By-Book Chapter Checklist

A. ORFANIDIS - ELECTROMAGNETIC WAVES AND ANTENNAS

Essential:

- Ch 1: Maxwell's Equations
- Ch 2: Uniform Plane Waves
- Ch 5: Reflection and Transmission
- Ch 7: Oblique Incidence

- Ch 11: Transmission Lines
- Ch 13: Impedance Matching
- Ch 14: S-Parameters
- Ch 16: Transmitting and Receiving Antennas
- Ch 21: Aperture Antennas, especially Microstrip Antennas if present in your edition
- Ch 22: Antenna Arrays

Useful/skimming:

- Ch 6: Multiple Dielectric Slabs
- Ch 9: Waveguides
- Ch 10: Resonant Cavities
- Ch 15: Radiation Fields
- Ch 18: Radiation from Apertures
- Ch 23: Array Design Methods

Skip initially:

- Ch 3: Pulse Propagation in Dispersive Media
- Ch 4: Propagation in Birefringent Media
- Ch 8: Multilayer Film Applications
- Ch 19: Diffraction
- Ch 20: Diffraction - Sommerfeld and Rayleigh-Sommerfeld theory
- Ch 24: Currents on Linear Antennas
- Ch 25: Coupled Antennas

B. POZAR - MICROWAVE ENGINEERING

Essential:

- Ch 1: Electromagnetic Theory review
- Ch 2: Transmission Line Theory
- Ch 3: Transmission Lines and Waveguides

Especially microstrip line, surface waves on grounded dielectric sheets, and waveguide basics.

- Ch 4: Microwave Network Analysis
- Ch 5: Impedance Matching and Tuning
- Ch 6: Microwave Resonators
- Ch 11: Active RF and Microwave Devices

Focus on diodes and device models.

- Ch 14: Introduction to Microwave Systems

Useful/skimming:

- Ch 7: Power Dividers and Directional Couplers
- Ch 8: Microwave Filters
- Ch 10: Noise and Nonlinear Distortion
- Ch 12: Microwave Amplifier Design
- Ch 13: Oscillators and Mixers

Skip initially:

- Ch 9: Theory and Design of Ferrimagnetic Components, unless a paper uses ferrite phase shifters.

C. BALANIS - ANTENNA THEORY: ANALYSIS AND DESIGN

Essential:

- Ch 2: Fundamental Parameters and Figures of Merit
- Ch 6: Arrays: Linear, Planar, and Circular
- Ch 12: Aperture Antennas, selected sections
- Ch 13: Horn Antennas, selected sections
- Ch 14: Microstrip Antennas
- Ch 15: Reflector Antennas

Useful/skimming:

- Ch 16: Smart Antennas, if your edition has it
- Ch 17: Antenna Measurements, if your edition has it

Less urgent:

- Ch 3: Radiation Integrals and Auxiliary Potential Functions
- Ch 4: Linear Wire Antennas
- Ch 5: Loop Antennas

D. GARG ET AL. - MICROSTRIP ANTENNA DESIGN HANDBOOK

Essential topics:

- Ch 1: Overview, advantages, limitations, feeding techniques, applications
- Rectangular microstrip patch antenna design
- Circular/other patch shapes, if needed
- Feeding methods
- Input impedance
- Matching
- Bandwidth
- Efficiency
- Arrays
- Reconfigurable/active microstrip antennas, if your edition has such sections

Note:

Chapter numbers may vary by edition or printing, so follow topic names.

E. NAYERI, YANG, ELSHERBENI - REFLECTARRAY ANTENNAS

Essential:

- Ch 1: Introduction
- Ch 2: Reflectarray Element Design
- Ch 3: Reflectarray System Analysis
- Ch 4: Reflectarray Radiation Pattern Calculation
- Ch 5: Reflectarray Bandwidth
- Ch 6: Reflectarray Design Examples

Useful:

- Ch 7: Broadband and Multiband Reflectarrays
- Ch 10: Beam Scanning Reflectarrays

- Ch 11: Reflectarray Applications

Why this book is important:
Reflectarrays are the classical antenna-engineering ancestor of many RIS/metasurface designs.

F. GOLDSMITH - WIRELESS COMMUNICATIONS

Essential:
- Ch 2: Path Loss and Shadowing
- Ch 3: Statistical Multipath Channel Models
- Ch 4: Capacity of Wireless Channels
- Ch 7: Diversity
- Ch 10: Multiple Antennas and Space-Time Communications
- Ch 12: Multicarrier Modulation
- Ch 15: Cellular Systems

Useful/skimming:
- Ch 5: Digital Modulation and Detection
- Ch 6: Performance of Digital Modulation over Wireless Channels

G. TSE AND VISWANATH - FUNDAMENTALS OF WIRELESS COMMUNICATION

Optional but strong supplement:
- Ch 2: The Wireless Channel
- Ch 3: Point-to-Point Communication
- Ch 5: Capacity of Wireless Channels
- Ch 7: MIMO I: Spatial Multiplexing and Channel Modeling
- Ch 8: MIMO II: Capacity and Multiplexing Architectures

Extra if needed:
- Ch 4: Cellular Systems
- Ch 10: MIMO Multiuser Communication

Use this book if you want a deeper theoretical understanding of MIMO and fading.

H. BOYD AND VANDENBERGHE - CONVEX OPTIMIZATION

Essential for RIS optimization papers:
- Ch 1: Introduction
- Ch 2: Convex Sets, basics only
- Ch 3: Convex Functions, basics only
- Ch 4: Convex Optimization Problems
- Ch 5: Duality, focus on Lagrangian and KKT
- Ch 6: Approximation and Fitting, focus on least squares
- Ch 9: Unconstrained Minimization, focus on descent, gradient, and Newton methods

Useful/skimming:
- Ch 11: Interior-Point Methods, only if papers mention SDP/SDR/barrier methods
- Appendix A: Mathematical background, especially linear algebra and probability if needed

I. TRETYAKOV - ANALYTICAL MODELING IN APPLIED ELECTROMAGNETICS

Advanced optional source:

- Ch 1: Introduction
- Ch 2: Thin Layers and Sheets
- Then read the sections/chapters on artificial impedance surfaces, periodic structures, arrays/meshes, and metamaterials/metasurfaces.

Use this after you have basic EM and microwave theory.

3. Your Reference List: Recommended Reading Order

FIRST RIS COMMUNICATION OVERVIEW:

1) [25] ElMossallamy et al. - Principles, challenges, opportunities
2) [4] Wu and Zhang - Towards smart and reconfigurable environment
3) [2] Basar et al. - Wireless communications through RIS
4) [3] Di Renzo et al. - Smart radio environments
5) [33] Li et al. - Intelligent metasurfaces: control, communication, computing

METASURFACE / METAMATERIAL BACKGROUND:

6) [16] Ali et al. - Metamaterials and metasurfaces review
7) [11] Glybovski et al. - Metasurfaces from microwaves to visible
8) [7] Kaina et al. - Binary tunable metasurfaces in reverberating media
9) [8] Liaskos et al. - Software-controlled metasurfaces
10) [12] Taravati and Eleftheriades - Space-time-modulated metasurfaces

REFLECTARRAYS / TUNABLE SURFACES:

11) [20] Hum et al. - Electronically tunable reflectarrays
12) [21] Sievenpiper et al. - Electrically tunable impedance surface
13) [22] Tian et al. - 1-bit digital reflective metasurface
14) [28] Kamoda et al. - 60-GHz 1-bit reflectarray
15) [29] Yang et al. - 1-bit multipolarization reflectarray element

LIGHT / IR / OPTICAL CONTROL:

16) [23] Light-controllable digital coding metasurfaces
17) [24] Light-controlled reflection phase modulation
18) [30] Infrared-controlled programmable metasurface
19) [31] Light-programmable metasurface
20) [32] IR communication system
21) [10] 2D-programmable RIS remotely controlled via digital infrared code

WIRELESS / MMWAVE / SURVEYS:

22) [1] Rappaport et al. - Millimeter wave mobile communications for 5G
23) [6] Survey on RIS below 10 GHz
24) [26] RIS-assisted non-terrestrial networks survey
27) [27] General RIS technology article

OPTIMIZATION / ML:

28) [9] Experiment-based deep learning approach for power allocation

REFERENCE / SUPPORTING MATERIAL:

29) [13] / [19] Microstrip Antenna Design Handbook
30) [14] Plasmonic metasurfaces, optical side
31) [15] Coupled-mode theory, advanced waveguide theory
32) [17] General metamaterials article
33) [18] Varactor diode overview

4. Key Concepts To Master

EM / microwave:

- Maxwell equations
- Plane wave
- Polarization
- Reflection coefficient
- Transmission coefficient
- Wave impedance
- Surface impedance
- Boundary conditions
- Transmission line
- S-parameters
- Smith chart
- Microstrip line
- Resonance

Antennas / arrays:

- Gain
- Directivity
- Radiation pattern
- Beamwidth
- Sidelobes
- Aperture efficiency
- Array factor
- Phase progression
- Beam steering
- Reflectarray
- Patch antenna

Metasurfaces:

- Unit cell
- Periodic structure
- Subwavelength element
- Reflection phase
- Reflection magnitude
- Phase quantization

- 1-bit / 2-bit coding
- Anomalous reflection
- Surface current
- Programmable metasurface

Wireless / RIS:

- Path loss
- Multipath
- Fading
- Channel matrix
- MIMO
- OFDM
- CSI
- Cascaded channel
- Passive beamforming
- SINR
- Capacity
- Near-field / far-field

Hardware:

- PIN diode
- Varactor diode
- Bias network
- RF choke
- DC blocking capacitor
- Phase shifter
- Switchable impedance
- Digital control
- IR control

Optimization:

- Objective function
- Constraint
- Convex / nonconvex problem
- Alternating optimization
- Gradient descent
- KKT conditions
- Least squares
- Semidefinite relaxation
- Discrete optimization

5. Practical Mini-Projects To Do While Reading

1) Array factor simulation

- Simulate a 1D array with N elements.
- Change the phase shift between elements.
- Observe beam steering.

2) Reflection coefficient calculation

- For a transmission line with a load impedance ZL, calculate Gamma.
- Plot magnitude and phase of Gamma.

3) Simple patch antenna calculation

- Choose a frequency, substrate permittivity, and substrate thickness.
- Estimate rectangular patch dimensions.

4) 1-bit RIS toy model

- Assume each RIS cell can reflect with phase 0 or 180 degrees.
- Simulate how different patterns steer the reflected beam.

5) RIS communication toy model

- Model received signal as direct channel + RIS-assisted channel.
- Vary the RIS phases to maximize received power.

6) Hardware reading task

- Pick one tunable metasurface paper.
- Identify what component tunes the phase: PIN diode, varactor, photodiode, MEMS, etc.
- Draw the bias/control path separately from the RF path.

6. What To Skip At First

Skip or only skim these until needed:

- Full rigorous derivation of diffraction theory
- Optical plasmonic metasurface details, unless your project is optical
- Coupled-mode theory of irregular waveguides, unless your supervisor specifically uses it
- Full convex optimization theory
- Full MIMO information theory proofs
- Full-wave simulation theory details
- Very advanced space-time-modulated metasurfaces

Your first target is not to become an EM theorist. Your first target is to understand what RIS papers mean when they say:

- reflection phase
- unit cell
- phase quantization
- beam steering
- cascaded channel
- passive beamforming
- channel estimation
- tunable impedance surface
- programmable metasurface

7. Minimal Path If You Are Short On Time

If you only have 3-4 weeks, do this:

1) Orfanidis:

- Ch 2, Ch 5, Ch 11, Ch 14, Ch 22

2) Pozar:

- Ch 2, Ch 3, Ch 4, Ch 5

3) Balanis:

- Ch 2, Ch 6, Ch 14

4) Nayeri Reflectarray Antennas:

- Ch 1, Ch 2, Ch 3

5) Goldsmith:

- Ch 2, Ch 3, Ch 10

6) Read papers:

- [25]
- [4]
- [2]
- [3]
- [11]
- [20]
- [21]
- [22]
- [10]

This minimal path will not make you an expert, but it should make the core references readable.

8. Recommended Note-Taking Template For Each Paper

For every paper, write one page with:

1) What problem does this paper solve?
2) Is it mainly about physics, hardware, communication theory, or optimization?
3) What is the system model?
4) What is the unit cell / surface / antenna structure?
5) What is being controlled?
   - phase?
   - amplitude?
   - polarization?
   - frequency?
   - beam direction?
6) What assumptions are made?
7) What equations are central?
8) What experiment or simulation validates it?
9) What are the limitations?
10) Which later reference builds on this?

9. Summary Of The Big Picture

RIS is not just one topic.

From the EM side:
RIS is a surface that modifies boundary conditions and controls scattering.

From the microwave hardware side:
RIS is an array of tunable resonant unit cells with controllable impedance/reflection phase.

From the antenna side:
RIS behaves like a passive/reconfigurable reflectarray or aperture.

From the wireless communication side:
RIS creates an additional controllable propagation path between transmitter and receiver.

From the optimization side:
RIS design means choosing many surface states to maximize communication performance under hardware constraints.

So the correct study path is not one book. It is a layered path:

EM waves -> RF/microwave networks -> antennas/arrays -> reflectarrays -> metasurfaces -> RIS channel models -> hardware control -> optimization.